

Continuous-time expansion for Kogut–Susskind: $SU(3)$ gapped for $g^2 \geq 388$, gap approaching $(8/3)g^2 \hbar c/a$

Yarotsky’s theorem gives the Kogut–Susskind gap for $SU(3)$ only beyond $g^2 \sim 10^{101}$, because its time discretization pays e^{64t_0} per perturbed site. The same statement with usable numbers comes from the Duhamel expansion of e^{-tH} in the plaquette term with the electric evolution kept exact between insertions: a polymer gas in continuous time whose weights carry the exact decay $e^{-E \Delta \tau}$ of every excited link, so no neighbourhood is ever left unpenalized. Three structural facts do the work. The electric term is diagonal in the character basis with eigenvalue $\varepsilon \sum_{\ell} C_2(r_{\ell})/C_2(f)$, $\varepsilon = (g^2/2)C_2(f) = 2g^2/3$. The plaquette operator $w_p = \text{tr } U_p + \text{tr } U_p^{\dagger}$ acts on four links with $\|w_p\| \leq 2N = 6$, and $V = (2/g^2) \sum_p (N - \text{Re tr } U_p)$ is $-(1/g^2) \sum_p w_p$ up to a constant. And Gauss’s law forces every intermediate excited set to be a closed flux loop, so at least four links are excited whenever any are, $E \geq 4\varepsilon$ on the orthogonal complement of the vacuum. The Kotecký–Preiss criterion, in its measure-space form, with the weight $a(\gamma) = n(\gamma) + \lambda |\text{supp } \gamma|$ and $\lambda = \varepsilon(1 - \theta)$, then gives a gap of at least $4\lambda = (8/3)g^2(1 - \theta)$ in units $\hbar c/a$ whenever

$$g^4 \geq \max\left(\frac{3132}{\theta}, \frac{783}{\sqrt{\theta(1-\theta)}}\right),$$

for the class of polymers in which each insertion touches the current excited set; so $\Delta_{\text{KS}} \geq \frac{4}{3}g^2 \hbar c/a$ for $g^2 \geq 79$, $\Delta_{\text{KS}} \geq 2.4g^2 \hbar c/a$ for $g^2 \geq 177$, and the bound approaches the flux-loop energy as $g \rightarrow \infty$, which fixes the constant γ of the T2 theorem at 4. The general polymers, in which a later insertion may start a separate piece that is bridged afterwards, are counted explicitly by a spanning tree of consecutive touchings, at a factor 24 per insertion, which makes the rigorous threshold $g^2 \geq 388$ for the same $\frac{4}{3}g^2$ gap. Constants explicit; nothing promoted.

1. The expansion

Units $\hbar c/a = 1$. Write $H = H_E - W + \text{const}$ with $H_E = (g^2/2) \sum_{\ell} (-\Delta_{\ell})$ and $W = (1/g^2) \sum_p w_p$. In the Peter–Weyl basis H_E is diagonal, its eigenvalue on a link configuration $\sigma = (r_{\ell})$ being $E(\sigma) = (g^2/2) \sum_{\ell} C_2(r_{\ell}) \geq \varepsilon |S(\sigma)|$, with $S(\sigma)$ the set of links carrying a nontrivial representation and $\varepsilon = (g^2/2)C_2(f)$, since $C_2(f) = 4/3$ is the least nonzero Casimir of $SU(3)$. Duhamel’s formula gives

$$e^{-tH} = \sum_{n \geq 0} \int_{0 < \tau_1 < \dots < \tau_n < t} e^{-(t-\tau_n)H_E} W e^{-(\tau_n-\tau_{n-1})H_E} \dots W e^{-\tau_1 H_E},$$

all terms positive. Insert, between consecutive factors, the resolution $1 = \sum_S P_S$ with P_S the projection onto configurations whose excited set is exactly S ; each P_S is a product over links. A **history** is then a sequence $(p_k, \tau_k, S_k)_{k=1}^n$ with $S_0 = \emptyset$, and its weight is

$$\left\langle \Omega_0, \prod_k (P_{S_k} \frac{1}{g^2} w_{p_k} P_{S_{k-1}}) e^{-\Delta \tau_k H_E} \Omega_0 \right\rangle, \quad \left| \cdot \right| \leq \left(\frac{2N}{g^2} \right)^n \prod_{k=1}^{n-1} e^{-\varepsilon |S_k| \Delta \tau_k},$$

because $\|w_p\| \leq 2N$ and $e^{-\Delta \tau H_E} P_S$ has norm at most $e^{-\varepsilon |S| \Delta \tau}$.

Two constraints on the S_k . First, $\text{links}(p_k) \subseteq S_{k-1} \cup S_k$: acting on a trivial link, w_p makes it nontrivial ($f \otimes 1 = f$), so a link of p_k can be trivial afterwards only if it was nontrivial before. Second, **Gauss’s law**: Ω_0 and every w_p are gauge invariant, so every intermediate configuration is, and the representations on the links at a vertex fuse to the trivial one; a nonempty excited set is therefore a union of closed flux loops and $|S_k| \geq 4$ whenever $S_k \neq \emptyset$. This is where the non-abelian theory pays nothing extra: the same constraint holds for $U(1)$.

Factorization. The support of a history is $\text{supp } \gamma = \bigcup_k S_k \times [\tau_k, \tau_{k+1}] \subset \mathcal{E} \times [0, t]$. If a history splits into pieces with disjoint supports, its weight is the product of the pieces’ weights,

because P_S , $e^{-\Delta\tau H_E}$ and Ω_0 factor over links and each w_p acts on four links; and the ordered time integral over the whole history is the product of the ordered integrals over the pieces. So

$$Z(t) = \langle \Omega_0, e^{-tH} \Omega_0 \rangle = \sum_{\{\gamma_i\} \text{ compatible}} \prod_i w(\gamma_i),$$

a gas of **polymers**, the connected histories, two polymers being compatible when their supports are disjoint. Between consecutive insertions of a connected polymer the excited set is nonempty, or the support would split in time; hence $|S_k| \geq 4$ for $1 \leq k \leq n-1$.

2. The convergence criterion

The Kotecký–Preiss criterion (Commun. Math. Phys. 103 (1986) 491) in its form for polymers indexed by a measure space (Poghosyan and Ueltschi, J. Math. Phys. 50 (2009) 053509; metadata level) is: if for every γ

$$\int_{\gamma' \not\sim \gamma} |w(\gamma')| e^{a(\gamma')} d\gamma' \leq a(\gamma),$$

then the cluster expansion of $\log Z$ converges absolutely and uniformly in the volume, and truncated correlations obey the standard bounds. Here $d\gamma'$ is the sum over shapes $(p_k, S_k)_{k=1}^n$ times $d\tau_1 \prod_{k=1}^{n-1} d\Delta\tau_k$.

Take

$$a(\gamma) = n(\gamma) + \lambda |\text{supp } \gamma|, \quad |\text{supp } \gamma| = \sum_{k=1}^{n-1} |S_k| \Delta\tau_k, \quad 0 < \lambda < \varepsilon.$$

Then $e^{a(\gamma')} |w(\gamma')| \leq (2Ne/g^2)^{n'} \prod_k e^{-(\varepsilon-\lambda)|S'_k| \Delta\tau'_k}$.

Overlap. $\gamma' \not\sim \gamma$ means some link ℓ is excited in both at a common time. For a fixed shape and internal times of γ' , the set of τ'_1 producing an overlap through link $\ell \in S'_k$ during the interval in which γ excites ℓ has measure at most $T_\ell(\gamma) + \Delta\tau'_k$, where $T_\ell(\gamma)$ is the total time γ keeps ℓ excited, and $\sum_\ell T_\ell(\gamma) = |\text{supp } \gamma|$.

Integrals. $\int_0^\infty e^{-(\varepsilon-\lambda)|S| \Delta\tau} d\Delta\tau = \frac{1}{(\varepsilon-\lambda)|S|}$ and $\int_0^\infty \Delta\tau e^{-(\varepsilon-\lambda)|S| \Delta\tau} d\Delta\tau = \frac{1}{(\varepsilon-\lambda)^2 |S|^2} \leq \frac{1}{4(\varepsilon-\lambda)^2} \cdot \frac{1}{|S|}$.

Collecting, with

$$Q_\lambda := \sum_{\text{shapes } \ni \ell_0} (n-1) \left(\frac{2Ne}{g^2} \right)^n \prod_{k=1}^{n-1} \frac{1}{(\varepsilon-\lambda)|S_k|}$$

the weighted number of polymer shapes through a fixed link, the left side of the criterion is at most $Q_\lambda |\text{supp } \gamma| + \frac{Q_\lambda}{4(\varepsilon-\lambda)} n(\gamma)$, and the criterion holds whenever

$$Q_\lambda \leq \lambda \quad \text{and} \quad Q_\lambda \leq 4(\varepsilon-\lambda).$$

3. Counting shapes

Adjacent growth. Consider first the shapes in which every insertion p_k , $k \geq 2$, shares a link with the current excited set S_{k-1} . The first plaquette contains ℓ_0 : in three dimensions a link lies in four plaquettes, so 4 choices, and $S_1 = \text{links}(p_1)$ is forced. At step $k \geq 2$ there are at most $4|S_{k-1}|$ plaquettes touching S_{k-1} and at most $2^4 = 16$ choices of S_k . Each step therefore contributes at most

$$\frac{2Ne}{g^2} \cdot \frac{1}{(\varepsilon-\lambda)|S_{k-1}|} \cdot 64|S_{k-1}| = \frac{128Ne}{g^2(\varepsilon-\lambda)} =: u,$$

the size $|S_{k-1}|$ cancelling between the entropy and the energy denominator, which is the mechanism that keeps the count geometric. So

$$Q_\lambda^{\text{adj}} \leq 4 \frac{2Ne}{g^2} \sum_{m \geq 1} m u^m = \frac{8Ne}{g^2} \frac{u}{(1-u)^2}.$$

General shapes, counted by a tree. A later insertion may start a piece disjoint from the current support that is bridged afterwards. To count all connected shapes, build the graph G' on the insertions in which, for every link ℓ and every maximal interval during which ℓ is excited, consecutive insertions touching ℓ are joined by an edge. Connectedness of the support is connectedness of G' , and the edges attached to a given link tile its excitation intervals without overlap, so for any spanning tree $T \subseteq G'$

$$\sum_{e \in T} |\Delta\tau_e| \leq |\text{supp } \gamma|, \quad \text{hence} \quad e^{-(\varepsilon-\lambda)|\text{supp } \gamma|} \leq \prod_{e \in T} e^{-(\varepsilon-\lambda)|\Delta\tau_e|}.$$

Each insertion has at most eight slots for G' -edges, the previous and next toucher of each of its four links, so T is a rooted tree with children in labelled slots: at most 4^n plane shapes times 8^{n-1} slot assignments. Given the parent and the slot, the child plaquette contains the slot's link and differs from the parent, at most 3 choices; the excited set after it, at most 16; the insertion factor $2Ne/g^2$; and the edge time integral $\int_0^\infty e^{-(\varepsilon-\lambda)\Delta\tau} d\Delta\tau = 1/(\varepsilon - \lambda)$, the time direction being fixed by the slot. The per-vertex factor is

$$u_{\text{gen}} = 32 \cdot 3 \cdot 16 \cdot \frac{2Ne}{g^2} \cdot \frac{1}{\varepsilon - \lambda} = \frac{3072Ne}{g^2(\varepsilon - \lambda)} = 24u,$$

and the sum over all connected shapes through a fixed link is bounded by the adjacent-growth sum with u replaced by $24u$; the overlap factor of Section 2 loses its $1/4$ because only one link's decay is used per edge, so the second condition reads $Q \leq \varepsilon - \lambda$. The loss against adjacent growth is the factor 24, coming from the tree entropy 32, the 3 plaquettes per slot and the single-link decay in place of $1/(4(\varepsilon - \lambda))$. Both counts are given below; the general one is rigorous as written, the adjacent one is what a careful count of the same shapes should approach.

4. Numbers for $SU(3)$

With $N = 3$, $\varepsilon = 2g^2/3$ and $\lambda = \varepsilon(1 - \theta)$,

$$u = \frac{128 \cdot 3e}{g^2 \cdot \theta \cdot 2g^2/3} = \frac{576e}{\theta g^4} = \frac{1566}{\theta g^4}.$$

Impose $u \leq \frac{1}{2}$ and, for the first condition of Section 2, $\frac{8Ne}{g^2} \cdot 4u \leq \lambda$, that is $96e \cdot \frac{1566}{\theta g^4} \leq \frac{2}{3}g^2(1 - \theta) \cdot g^2/g^2$; cleaning up,

$$g^4 \geq \frac{3132}{\theta} \quad \text{and} \quad g^8 \geq \frac{6.13 \times 10^5}{\theta(1 - \theta)},$$

the second condition of Section 2 being weaker. Hence:

θ	threshold g^2	gap $\geq 4\lambda = \frac{8}{3}g^2(1 - \theta)$
1/2	79	$\frac{4}{3}g^2$
1/4	112	$2g^2$
1/10	177	$2.4g^2$
$\theta \rightarrow 0$	$g^2 \gtrsim \sqrt{3132/\theta}$	$\rightarrow \frac{8}{3}g^2$

With the general count, $u_{\text{gen}} = 37584/(\theta g^4)$, the conditions $u_{\text{gen}} \leq \frac{1}{2}$ and $4Fu_{\text{gen}} \leq \lambda$ give $g^4 \geq 75168/\theta$ and $g^8 \geq 1.47 \times 10^7/(\theta(1 - \theta))$:

θ	rigorous threshold g^2	gap $\geq \frac{8}{3}g^2(1-\theta)$
1/2	388	$\frac{4}{3}g^2$
1/4	548	$2g^2$
1/10	867	$2.4g^2$

The gap here is read off as in the Wilson note §3: running the criterion with $a(\gamma) = n + \lambda|\text{supp } \gamma|$ puts a factor $e^{-\lambda|\text{supp } X|}$ on every cluster, a cluster spanning a time interval of length t has $|\text{supp}| \geq 4t$ by Gauss's law, so vacuum-subtracted correlations in Euclidean time decay at rate at least 4λ , and for a positive semigroup that is a lower bound on the gap on the cyclic subspace of local observables, which in finite volume is everything by T1.

Consequence for T2. The T2 theorem's constant γ , defined by $\Delta \geq \gamma(g^2/2)C_2$, is $4(1-\theta)$ in this expansion and tends to 4: the gap is the energy of one flux loop, four links in the fundamental, up to corrections of relative order $1/g^4$. This agrees with the two-sided strong-coupling statement of the $SU(3)$ constants §2, whose upper side is $12g^2$, and closes the ratio $18/\gamma$ there to 4.5.

5. Why this beats the time-discretized argument

Yarotsky's Lemma 3 gives up the decay of every link in the range of a perturbed site for a whole step t_0 , and $t_0 \geq 7$ is forced by the entropy of the space-time lattice; the product is $e^{64t_0} \sim e^{450}$ per insertion. In continuous time each excited link decays exactly for exactly as long as it is excited, the insertion carries no step, and the only entropy is the choice of the next plaquette and excited set, which the energy denominator $1/((\varepsilon - \lambda)|S_k|)$ pays for link by link. The threshold drops from 10^{101} to 10^2 for the same model and the same conclusion.

6. Consequence for STATE

The T2 theorem for the Kogut–Susskind Hamiltonian now has an explicit threshold, $g^2 \geq 388$ for $SU(3)$ rigorously and $g^2 \geq 79$ in the adjacent-growth class, and an explicit gap approaching the flux-loop energy $(8/3)g^2 \hbar c/a$. Together with the Wilson transfer-matrix result, the strong-coupling boundary of the intermediate region is of order $g^2 \sim 10^2$ for both standard regularizations. The weak side remains the crude $1/g^2 \gtrsim 10^2$, and the intermediate region between them is unchanged.